

Trainings ▾

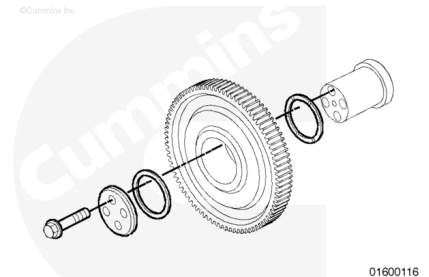
Install

The grooves in the thrust washer **must** be positioned toward the gear.

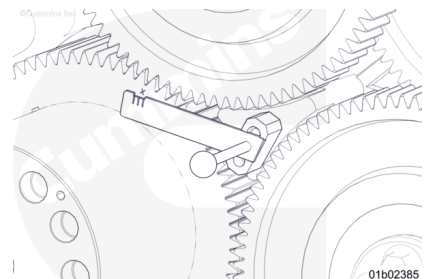
The timing marks on the camshaft idler gear **must** be visible when the gear is installed.

Use Lubriplate™ 105, or equivalent to lubricate the gear bushing, shaft, and thrust washer.

Lubricate the capscrews with clean engine oil. Assemble the parts as illustrated.



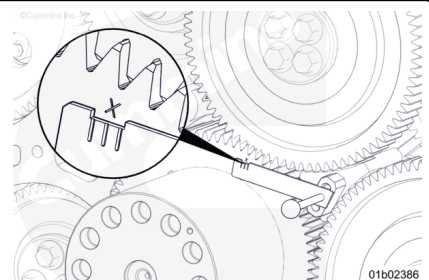
Insert the crankshaft timing service tool, Part Number 5394520, into the position shown in the front gear housing.



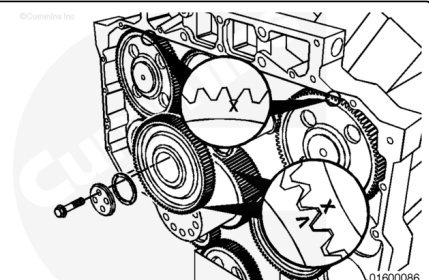
Use the crankshaft timing service tool to set the crankshaft timing mark at the Top Dead Centre (TDC) position.

Rotate the crankshaft so that the 'X' timing mark on the crankshaft gear aligns as closely as possible to the center line on the crankshaft timing tool.

Note : The 'X' timing mark on the crankshaft gear **must** fall between the two outer lines on the crankshaft timing tool.



Align the "V" on the idler gear with the "X" on the crankshaft gear. Align the "X" on the camshaft gear at 12 o'clock and with the dowel holes in the front gear housing.



Rotate the idler shaft as necessary to align the mounting capscrew holes.

The oil holes in the idler shaft **must** be placed in the vertical position.

Use the capscrews to pull the shaft into the bore.

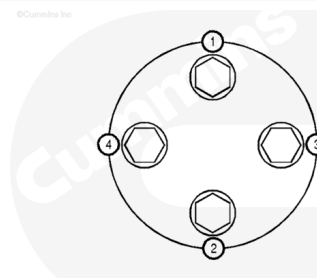
Using the following steps, tighten the capscrews in the sequence shown.

Torque Value:

1. 80 n•m [59 ft-lb]

2. 165 n•m [122 ft-lb]

3. 280 n•m [207 ft-lb]

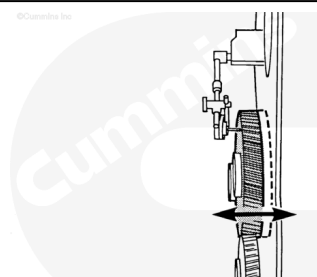


01600146

Measure the camshaft idler gear end clearance.

Left Bank Camshaft Idler Gear End Clearance

mm		in
0.15	MIN	0.006
0.33	MAX	0.013



if400ta